Multivariate Central Limit Theorem & Chi-Square Tests

Lecture 11

1 Multivariate Central Limit Theorem (MCLT) and Quadratic Forms

1.1 Quadratic Form Representation

We analyze the quadratic form in the exponent of the multivariate distribution. Let the quadratic form be defined as:

$$x^T Ax = \sum_i \sum_j x_i x_j A_{ij} \tag{1}$$

1.2 Derivation of the Chi-Square Statistic

We aim to show the algebraic identity that relates the covariance structure to the standard Chi-square sum. Let the statistic B_n be defined involving the inverse covariance entries $(\frac{1}{p_i}\delta_{ij} + \frac{1}{p_k})$,

$$\delta_{ij} = I(i = j)$$

:

$$B_n = n \sum_{i=1}^{k-1} \sum_{j=1}^{k-1} \left(\frac{X_i}{n} - p_i \right) \left(\frac{X_j}{n} - p_j \right) \left(\frac{1}{p_i}\delta_{ij} + \frac{1}{p_k} \right) \tag{2}$$

Expanding the terms inside the summation:

$$B_n = \sum_{i=1}^{k-1} \sum_{j=1}^{k-1} (X_i - np_i)(X_j - np_j) \frac{1}{np_k} + \sum_{i=1}^{k-1} \frac{(X_i - np_i)^2}{np_i} \tag{3}$$

$$= \frac{1}{np_k} \sum_{i=1}^{k-1} (X_i - np_i) \left[\sum_{j=1}^{k-1} (X_j - np_j) \right] + \sum_{i=1}^{k-1} \frac{(X_i - np_i)^2}{np_i} \tag{4}$$

Recall the constraint $\sum_{i=1}^{k} (X_i - np_i) = 0$, which implies:

$$\sum_{j=1}^{k-1} (X_j - np_j) = -(X_k - np_k) \tag{5}$$

Substituting this back into the equation:

$$\text{Cross Term} = \frac{1}{np_k} \left[-(X_k - np_k) \right] \left[-(X_k - np_k) \right] \tag{6}$$

$$= \frac{(X_k - np_k)^2}{np_k} \tag{7}$$

Thus, the full summation becomes the standard Pearson goodness-of-fit statistic:

$$B_n = \sum_{i=1}^{k-1} \frac{(X_i - np_i)^2}{np_i} + \frac{(X_k - np_k)^2}{np_k} = \sum_{i=1}^{k} \frac{(X_i - np_i)^2}{np_i} \tag{8}$$

2 Goodness of Fit Test

2.1 Hypothesis and Statistic

We test if the data follows a specified distribution p_0:

- $H_0 : p = p_0$
- $H_a : p \neq p_0$
- test for goodness of fit

The test statistic is:

$$B_n = \sum_{i=1}^{k} \frac{(X_i - np_{i0})^2}{np_{i0}} \tag{9}$$

2.2 Asymptotic Distribution

Under H_0, as $n \to \infty$:

$$B_n \longrightarrow \chi^2_{k-1} \tag{10}$$

Decision Rule: Reject H_0 for large values of B_n.

3 Test of Independence

3.1 Two-Way Contingency Table

Consider two categorical variables X and Y:

- $X \in \{x_1, \ldots, x_I\}$ (Rows)
- $Y \in \{y_1, \ldots, y_J\}$ (Columns)

Let

$$n_{ij} = \text{number of observations with } X = x_i, Y = y_j$$

be the observed count in the cell (i, j). The total sample size is $n = \sum_{i,j} n_{ij}$.

3.2 Independence Assumption

If X and Y are independent:

$$P(X = x_i \cap Y = y_j) = P(X = x_i)P(Y = y_j) \quad \forall i \in 1, 2..., I, j \in 1, 2, 3..., J \tag{11}$$

3.3 Hypothesis Testing

- $H_0 : p_{ij} = p_{i\cdot}p_{\cdot j}$ (Independence)
- H_a: Variables are not independent.

The underlying distribution is Multinomial $\sim \text{Mult}(n, \{p_{ij}\})$.

3.4 Expected Counts and Statistic

The statistic is defined as:

$$B = \sum_{i=1}^{I}\sum_{j=1}^{J} \frac{(n_{ij} - E_{ij})^2}{E_{ij}} \tag{12}$$

Where the expected count E_{ij} under H_0 is estimated using marginal proportions:

$$\hat{p}_{i\cdot} = \frac{n_{i\cdot}}{n}, \quad \hat{p}_{\cdot j} = \frac{n_{\cdot j}}{n} \tag{13}$$

$$E_{ij} = n \times (\hat{p}_{i\cdot} \times \hat{p}_{\cdot j}) = \frac{n_{i\cdot} n_{\cdot j}}{n} \tag{14}$$

3.5 Degrees of Freedom (df)

The degrees of freedom are calculated by subtracting the number of estimated parameters from the total independent cells:

$$\begin{aligned}
\text{df} &= (\text{Params in full model}) - (\text{Params under } H_0) && (15)\\
&= (IJ - 1) - [(I-1) + (J-1)] && (16)\\
&= IJ - 1 - I + 1 - J + 1 && (17)\\
&= I(J-1) - (J-1) && (18)\\
&= (I-1)(J-1) && (19)
\end{aligned}$$

Historical Note:

- Pearson originally proposed $df = IJ - 1$.
- Fisher corrected this in 1922 to $(I-1)(J-1)$.
- These tests are asymptotic, need large n.

4 Approximation Guidelines

The Chi-square test is an asymptotic test.

- **For Binomial (n,p):** If testing $H_0 : p = p_0$:

$$Z = \frac{X/n - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \overset{\text{approx}}{\sim} N(0,1) \tag{20}$$

- **Rule of Thumb for n:**
  1. If $p_0 \approx 1/2$, even $n = 10$ gives a good approximation.
  2. If p_0 is close to 0 or 1, we need $n \geq 30$.
  3. Class frequencies (np,n(1-p)) should be large.
- **For Chi-Square (Multinomial):** In goodness of fit or test of independence (generalisation of binomial to multinomial) a rule of thumb is each observed frequency ≥ 5
- Two way stands for 2 variables X and Y. If there are more variables , this can be used to test pairwise independence.

5 Test for Homogeneity

5.1 Concept

- **Context:** We have a fixed number of levels (say I) for each level of a predictor variable X the response Y$\sim Mult_j(..., p_i)$.
- **Response:** A categorical variable Y (e.g., Cured, Improved, No Change).
- **Hypothesis H_0:** The distribution of Y is the same (homogeneous) across all levels of X.

$$p_{1j} = p_{2j} = \cdots = p_{Ij} \quad \forall j \tag{21}$$

Mathematically, the calculation is identical to the Test of Independence, but the sampling scheme (fixed row totals) differs.

5.2 Example: Aspirin & Heart Attack (1988)

Group	**Fatal Attack**	**Non-fatal**	**No attack**	**Total**
Placebo (x1)	18	171	10,845	11,034
Aspirin (x2)	5	99	10,933	11,037
Total	23	270	21,778	22,071

Table 1: Contingency Table for Homogeneity Test

Hypothesis for this Example:

- H_o :Conditional distribution of Y given $X = x_i$ is same for all i.
- H_0: $p_{\text{placebo}} = p_{\text{aspirin}}$ (The treatment and control groups have the same distribution of outcomes).
- H_a: The distributions differs / Not Ho.

$$\beta_{ij} = P(X = i \text{ and } Y = j) = P(X = i) \cdot P(Y = j \mid X = i) = p_{ii} \cdot p_{ij} \quad \text{under } H_0.$$

This is same as test of independence mathematically.